JAVA PRACTICAL PROGRAMS

19 Short Programs with Outputs and Viva Questions

Record-note style • Fixed values • Easy revision

Study tip: Memorize the common main() structure, then practise the loop and OOP patterns.

1. Even or Odd

Program

```
import java.io.*;
class evenodd
{
    public static void main(String args[])
    {
        int n=10;
        if(n%2==0)
            System.out.println("Even");
        else
            System.out.println("Odd");
    }
}
```

Output

```
Even
```

Viva Questions and Answers

- **What is an even number?** A number divisible by 2.
- **What is %?** The modulus operator; it gives the remainder.
- **What is if-else?** A conditional statement.

2. Multiplication Table

Program

```
import java.io.*;
class table
{
    public static void main(String args[])
    {
        int i,n=5;
        for(i=1;i<=10;i++)
        {
            System.out.println(n*i);
        }
    }
}
```

Output

```
5\n10\n15\n20\n25\n30\n35\n40\n45\n50
```

Viva Questions and Answers

- **What is a loop?** A statement used to repeat instructions.
- **What is a for loop?** A loop used to repeat code.
- **What is i++?** It increases i by 1.

3. Largest Among Three Numbers

Program

```
import java.io.*;
class largest
{
    public static void main(String args[])
    {
        int a=10,b=20,c=15;
        if(a>b && a>c)
            System.out.println(a);
        else if(b>c)
            System.out.println(b);
        else
            System.out.println(c);
    }
```

```
}
```

Output

```
20
```

Viva Questions and Answers

- **What is &&?** The logical AND operator.
- **What is else if?** It checks another condition when the previous condition is false.
- **What does > mean?** Greater-than operator.

4. Sum and Average of Array

Program

```
import java.io.*;
class array
{
    public static void main(String args[])
    {
        int a[]={10,20,30};
        int sum=0,i;
        for(i=0;i<3;i++)
            sum=sum+a[i];

        System.out.println("Sum="+sum);
        System.out.println("Average="+sum/3.0);
    }
}
```

Output

```
Sum=60\nAverage=20.0
```

Viva Questions and Answers

- **What is an array?** A collection of elements of the same type.
- **What is the first index of an array?** 0.
- **How do you calculate average?** Sum divided by the number of elements.

5. Fibonacci Series

Program

```
import java.io.*;
class fibonacci
{
    public static void main(String args[])
    {
        int a=0,b=1,c,i;
        for(i=1;i<=7;i++)
        {
            System.out.print(a+" ");
            c=a+b;
            a=b;
            b=c;
        }
    }
}
```

Output

```
0 1 1 2 3 5 8
```

Viva Questions and Answers

- **What is the Fibonacci series?** Each term is the sum of the previous two terms.
- **What are the first two terms?** 0 and 1.
- **Which loop is used?** for loop.

6. String Palindrome

Program

```
import java.io.*;
class palindrome
{
    public static void main(String args[])
    {
        String s="madam",r="";
        int i;
        for(i=s.length()-1;i>=0;i--)
            r=r+s.charAt(i);

        if(s.equals(r))
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```

Output

```
Palindrome
```

Viva Questions and Answers

- **What is a palindrome?** A word that reads the same forward and backward.
- **What does length() do?** Returns the length of a string.
- **What does charAt() do?** Returns the character at a specified index.
- **What does equals() do?** Compares the contents of strings.

7. Display 1 to 100 Using do-while

Program

```
import java.io.*;
class dowhile
{
    public static void main(String args[])
    {
        int i=1;
        do
        {
            System.out.println(i);
            i++;
        }
        while(i<=100);
    }
}
```

Output

```
1\n2\n3\n...\n100
```

Viva Questions and Answers

- **What is a do-while loop?** It checks the condition after executing the loop body.
- **Does it execute at least once?** Yes.
- **What does i++ do?** It increases i by 1.

8. Factorial Using for Loop

Program

```
import java.io.*;
class factorial
{
    public static void main(String args[])
    {
        int n=5,f=1,i;
```

```
        for(i=1;i<=n;i++)
            f=f*i;

        System.out.println("Factorial="+f);
    }
}
```

Output

```
Factorial=120
```

Viva Questions and Answers

- **What is factorial?** The product of integers from 1 to a given positive number.
- **What is 5!?** 120.
- **What is 0!?** 1.

9. Even Numbers from 1 to 50

Program

```
import java.io.*;
class even
{
    public static void main(String args[])
    {
        int i;
        for(i=2;i<=50;i=i+2)
            System.out.println(i);
    }
}
```

Output

```
2\n4\n6\n...\n50
```

Viva Questions and Answers

- **Why does the loop start at 2?** 2 is the first positive even number.
- **What does i=i+2 do?** It increases i by 2.
- **How many even numbers are printed?** 25.

10. Uppercase and Lowercase

Program

```
import java.io.*;
class stringcase
{
    public static void main(String args[])
    {
        String s="hello java";
        System.out.println(s.toUpperCase());
        System.out.println(s.toLowerCase());
    }
}
```

Output

```
HELLO JAVA\nhello java
```

Viva Questions and Answers

- **What does toUpperCase() do?** Converts letters to uppercase.
- **What does toLowerCase() do?** Converts letters to lowercase.
- **Is String a class in Java?** Yes.

11. Single Inheritance

Program

```
import java.io.*;
```

```
class A
{
    void show()
    {
        System.out.println("Hello");
    }
}
class B extends A
{
    public static void main(String args[])
    {
        B obj=new B();
        obj.show();
    }
}
```

Output

```
Hello
```

Viva Questions and Answers

- **What is inheritance?** Acquiring methods and properties from another class.
- **What is single inheritance?** One child class inherits from one parent class.
- **Which keyword is used?** extends.

12. Multilevel Inheritance

Program

```
import java.io.*;
class A
{
    void show()
    {
        System.out.println("A");
    }
}
class B extends A
{
}
class C extends B
{
    public static void main(String args[])
    {
        C obj=new C();
        obj.show();
    }
}
```

Output

```
A
```

Viva Questions and Answers

- **What is multilevel inheritance?** Inheritance through multiple levels of classes.
- **Which keyword is used?** extends.
- **Can class C access class A's method?** Yes, through inheritance.

13. Abstract Class

Program

```
import java.io.*;
abstract class A
{
    abstract void show();
}
class B extends A
{
```

```
    void show()
    {
        System.out.println("Hello");
    }
    public static void main(String args[])
    {
        B obj=new B();
        obj.show();
    }
}
```

Output

```
Hello
```

Viva Questions and Answers

- **What is an abstract class?** A class declared using the abstract keyword.
- **Can we create its object directly?** No.
- **What is an abstract method?** A method declared without a body.

14. Interface

Program

```
import java.io.*;
interface A
{
    void show();
}
class B implements A
{
    public void show()
    {
        System.out.println("Hello");
    }
    public static void main(String args[])
    {
        B obj=new B();
        obj.show();
    }
}
```

Output

```
Hello
```

Viva Questions and Answers

- **What is an interface?** A reference type that defines a contract for classes.
- **Which keyword is used?** implements.
- **Can a class implement multiple interfaces?** Yes.

15. Thread Class

Program

```
import java.io.*;
class A extends Thread
{
    public void run()
    {
        System.out.println("Thread");
    }
    public static void main(String args[])
    {
        A obj=new A();
        obj.start();
    }
}
```

Output

```
Thread
```

Viva Questions and Answers

- **What is a thread?** A unit of execution.
- **Which method contains the task?** run().
- **Which method starts a new thread?** start().

16. Runnable Interface

Program

```
import java.io.*;
class A implements Runnable
{
    public void run()
    {
        System.out.println("Thread");
    }
    public static void main(String args[])
    {
        A obj=new A();
        Thread t=new Thread(obj);
        t.start();
    }
}
```

Output

```
Thread
```

Viva Questions and Answers

- **What is Runnable?** An interface used to define a thread's task.
- **Which method must be implemented?** run().
- **Which method starts the thread?** start().

17. Exception Handling

Program

```
import java.io.*;
class exceptiondemo
{
    public static void main(String args[])
    {
        try
        {
            int a=10/0;
        }
        catch(Exception e)
        {
            System.out.println("Error");
        }
    }
}
```

Output

```
Error
```

Viva Questions and Answers

- **What is an exception?** An abnormal condition during program execution.
- **What is try?** It contains code that may cause an exception.
- **What is catch?** It handles an exception.
- **Which exception occurs when integer division by zero is attempted?** ArithmeticException.

18. Reading a File

Program

```
import java.io.*;
class readfile
{
    public static void main(String args[]) throws Exception
    {
        FileReader f=new FileReader("a.txt");
        int i;
        while((i=f.read())!=-1)
            System.out.print((char)i);
        f.close();
    }
}
```

Output

```
Hello (if a.txt contains Hello)
```

Viva Questions and Answers

- **What is FileReader?** A class used to read character files.
- **What does read() return at the end of a file?** -1.
- **Why is close() used?** To release the file resource.

19. GUI Login Form

Program

```
import javax.swing.*;
class login
{
    public static void main(String args[])
    {
        JFrame f=new JFrame("Login");
        JButton b=new JButton("Login");
        b.setBounds(50,50,100,30);
        f.add(b);
        f.setSize(250,150);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

Output

```
A window appears with a Login button.
```

Viva Questions and Answers

- **What is GUI?** Graphical User Interface.
- **What is Swing?** A Java GUI toolkit.
- **What is JFrame?** A top-level window.
- **What is JButton?** A button component.

Common Java Viva Questions

- **What is Java?** An object-oriented programming language.
- **Who developed Java?** James Gosling and his team at Sun Microsystems.
- **What is JDK?** Java Development Kit.
- **What is JVM?** Java Virtual Machine.
- **What is a class?** A blueprint for creating objects.
- **What is an object?** An instance of a class.
- **What is a method?** A block of code that performs a task.
- **What is a constructor?** A special member used to initialize objects.
- **What is a variable?** A named storage location for a value.
- **Is Java platform-independent?** Yes, Java bytecode can run on a compatible JVM.